

# Reading the World Through Data: Bubble Charts

Gapminder Tools · Gapminder

Years 7-10

~30 minutes

Mathematics · Statistics & Probability

## NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://www.gapminder.org/tools/>

Curriculum links: AC9M9ST04 · AC9S10I05 · AC9S7I05 · AC9S8I05

## Learning intentions

- I can select variables to compare using an animated bubble chart.
- I can describe trends, relationships and outliers I observe as data changes over time.
- I can justify why a bubble or scatter display is a useful way to show this kind of data.

## Getting started

Open Gapminder Tools and choose two variables for your axes (e.g. income per person and life expectancy). Press play and watch how the bubbles (countries) move across the chart from the earliest year shown to the most recent.

## Tasks

1. Before pressing play, look at the chart in the earliest year shown. Name one country that stands out (e.g. an outlier) and describe why.

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2. Press play and watch the animation run to the present day. Describe the general trend you see — do most bubbles move up, down, or sideways over time?

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3. Pick one country and follow its bubble through the whole animation. Describe how its position changed and suggest one real-world event or trend that might explain this.

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4. Pause the animation at three different years (an early year, a middle year, a recent year) and record the approximate values for your chosen country.

| Year | Variable 1 value | Variable 2 value |
|------|------------------|------------------|
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|      |                  |                  |
|      |                  |                  |

5. Change one of your axis variables to a new one of your choice. Does the overall pattern of countries change? Describe what's different.

6. Explain why an animated bubble chart, rather than a single static graph, is a good way to show how these variables change over time and across many countries.

7. Identify one anomaly (a country that doesn't follow the general trend) in your chart. Suggest a possible reason for this anomaly.

**Extension (optional)**

Choose two variables you think might NOT be related (no obvious connection). Test your prediction in Gapminder and explain whether the data supported or challenged your assumption.